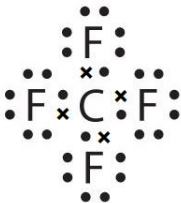


Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		ionic	1			1		
		(ii)		award (1) for either of following melted / made molten dissolved - accept put into water neutral answer - heated ions are free to move (1) do not accept reference to electrons being free to move	2			2		
	(b)	(i)		award (2) for correct diagram  accept any representation of electrons e.g. all crosses or all dots if incorrect award (1) for all four shared pairs		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		(individual) molecules are not joined / weak forces of attraction between molecules (1) little energy/heat needed to break (intermolecular) forces (of attraction) (1)	2			2		
	(c)			absorbs UV rays / reflects UV rays / transparent (1) neutral answer – absorbs sunlight catalyses the breakdown of dirt / enables water to spread out (into a film) on the window / hydrophilic (1)	2			2		
				Question 3 total	7	2	0	9	0	0